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PAN Card Tampering Detection using Machine Learning Techniques

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***Abstract*— In an era dominated by digital transactions and an increasing reliance on electronic identity verification, the need for robust security measures has become important. As people and organizations increasingly undertake financial transactions, deal with sensitive data, and perform numerous online activities, the vulnerability to cybersecurity risks and identity fraud has become a serious concern. The machine learning-based PAN card tampering detection system offers an innovative solution to address the growing issue of identity fraud and document manipulation, particularly within the financial and government sectors. The dataset consists of 40 real and 40 fake PAN cards and also 20 real and 9 fake Aadhar cards. The system uses CNNs, a specific type of deep learning model intended for image processing, to mitigate this problem. The performance of CNN is also compared against various pre-trained models like ResNet50, VGG16, Xception and MobileNetV2. After thorough analysis, this study concluded that CNN is the best performing model with an accuracy of 100%, loss 0.14% and average execution time of 0.24%. It meticulouslyanalyzes any discrepancies and variations such as alterations intext, wrong numerical data and tampering with the photos. The incorporation of CNN into this novel system not only resolves the existing vulnerabilities related to PAN cards but also lays the groundwork for an identity verification environment that is both more secure and reliable in the context of financial and governmental transactions.**

***Keywords*— PAN (Permanent Account Number) Card, Aadhar card, CNN (Convolutional Neural Network), forgery, identity fraud, security.**

I. Introduction

One critical aspect of this security landscape is the safeguarding of identity documents such as PAN (Permanent Account Number) cards and Aadhar cards against tampering and fraudulent activities. Conventional approaches to document verification are proving insufficient in combating the constantly evolving tactics employed by fraudsters. The inadequacy of reactive strategies urges the need for a proactive and technologically sophisticated solution tailored to address the nuanced challenges of document security, especially concerning PAN cards. The absence of such a comprehensive system exacerbates the inherent risks associated with tampering, necessitating the development of a dynamic framework. In India, the significance of the PAN card as a pivotal element in financial transactions, banking verifications, corporate processes, and government services is indisputable. This necessitates the development of a comprehensive and dynamic system that not only creates a diverse dataset for training and validation but also optimizes preprocessing for accurate tampering identification. Hence, CNN will be employed by this system. The machine learning-based PAN card tampering detection system is a significant step forward in identity verification. This system, with the aid of CNNs, provides a strong solution to the existing gaps in identity verification. Such an approach provides efficient security for the PAN cards. CNN is a specialized neural network designed to process images using exceptional accuracy. A significance of such networks is their ability to autonomously detect the features at large scale. This feature removes the requirement for manual feature engineering.This is an increased level of efficiency and enables the CNNsto detect patterns and features from data regardless of location, orientation, size, or translation.

CNNs follow a specific architecture, having several key components that will replace the human brain's capacity in recognizing patterns and features in images. Convolutional layers, pooling layers, fully connected layers, and ReLU activation functions are some of these constituents. Convolutional layers generate the image input by use of filters or kernels. Such layers enable the network to detect specific patterns or images such as lines, curves, or shapes. The model becomes non-linear when the ReLU activation function is used. The model can now learn more intricate patterns thanks to it. The input's spatial dimension is reduced using pooling layers. This helps to lower the model's computing cost and hence avoid overfitting. Finally, fully connected layers allow the network to use the observed features to forecast the input.

The integration of CNNs into the PAN card tampering detection system provides a better position to critically analyze the PAN card and Aadhar card images. Byidentifying subtle signs of tampering or forgery, it significantly enhances the security and reliability of identity verification procedures. Such innovative implementation of the CNNs not only upgrades the current vulnerabilities of the PAN cards but also paves the way for a more secure and trustworthy ecosystem in identity verification for financial and governmental transactions. Furthermore, recognizing theimperative role of seamless identity verification in various sectors, the system extends its capabilities by providing software for Know Your Customer (KYC) processes. This system enhances the efficiency and security of identity verification across sectors, offering a streamlined and standardized approach to KYC procedures.

There are 5 sections in this research study. Section II describes the literature survey of the existing methods and comparing their objectives, proposed systems, implemented methodologies and limitations. Section III describes the proposed system and the steps to implement the system. Section IV discusses about

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the different algorithms used for testing the system, which are CNN, Xception, ResNet50, VGG16 and MobileNetV2. The final section, Section V discusses about the experimental results and an analysis of the best performing model among the five. The last section concludes this research study.

II. RELATED WORKS

This research study highlights the challenges associated with identity fraud and document tampering, emphasizing the vulnerabilities inherent in traditional methods of document verification. Notably, the utilization of machine learning, specifically CNN has gained prominence in addressing similar challenges in various domains.

The work in [1] highlights the critical role of feature selection in fraud detection techniques of credit card using genetic algorithms to optimize feature selection and SMOTE to address the issue of data imbalance. It also does not address techniques to adapt to evolving patterns. Fuzzy membership functions are proposed as a solution to the data imbalance issue in [2] and the privacy concerns around the use of credit card transaction data are not discussed in detail.The research in [3] allows several parties to cooperatively train machine learning models without disclosing their private information to one another. It also addresses the dataprivacy concerns by utilizing encryption techniques such as SMPC and homomorphic encryption. Privacy preservation is not discussed about in [1] and [2]. In [4], the study utilizes decision tree classifiers as one of the primary methods and addresses privacy and security concerns by suggesting strategies such as data anonymization and secure data handling.

In [5], researchers focused on fraud detection in credit cards using Local outlier factor and Isolation Forest algorithms and the result of this research indicates an accuracy of 99.6% but witha low precision of 28-33% due to imbalance between valid and fraudulent transactions. In [6], to handle the credit card dataset imbalance issue, Matthew's correlation coefficient is identified as a strong measure for this issue and the SMOTE method helps in improving the performance of the classifiers.In [7], blockchain networks can be used to enhance security when it comes to credit card transactions and employs SHA 256 to secure data at each stage and uses Random Forest (RF)to optimize model accuracy. The study in [8] employs ANN for fraud detection in credit cards as it outperforms other models like SVM and KNN in terms of performance metricslike accuracy, precision and recall.

In [9], the study shows the usefulness of RF in image fraud detection as it overcomes the risk of overfitting and achieves an accuracy of 90% in classifying fake cards. The drawbacks of SVM, cluster analysis, and Naive Bayes classifiers are also addressed. In [10], after comparing the performance of various models to detect credit card fraud, SVM, KNN and CNN excelled individually using the European dataset, ensemble methods combining KNN, SVM, and Random Forest yield improved MCC and AUC using the Australian dataset and SVM, Random Forest, and CNN excelled individually in the German dataset [16]. CNNs outperform other DL methods and Autoencoders provide a solution for dynamic fraud patterns [17]. In [11], researchers discussed about detecting copy-move forgery using CMFD approaches like block based and key point-based methods. SIFT and SURF are some of the techniques used in CMFD where SIFT is robust against scaling and SURF has faster matching speeds.In [12], the study discussed about the usage of LBP (Local Binary pattern) to approach a texture-based analysis of document forgery as LBP descriptors are computed from image patches around text regions to capture texture discontinuities indicative of manipulation. It achieved a forgery detection rate of 7.38% and a low false positive ratio of 0.05%.

In [13], researchers have proposed a new method to distinguish documents produced by laser printers and inkjet printers using SVM by separating individual characters and determining the origin ofthe document based on the aggregate vote from decisions onindividual characters. The accuracy reached 90% for inkjet printers and most laser printers and copiers and was robust for JPEG compression. In [14], have proposed a copy-move forgery detection technique utilizing AKAZE features and RANSAC for eliminating false matches and the robustness outperformed SIFT and SURF based on manipulators like white gaussian noise, rotation and JPEG compression. In [15], have presented how to detect duplicated regions in natural images using DWT which can only detect translationduplicate forgery and KPCA which can detect both translation-flip and translation-rotation duplications and also achieves the highest true positive rates with false positives.

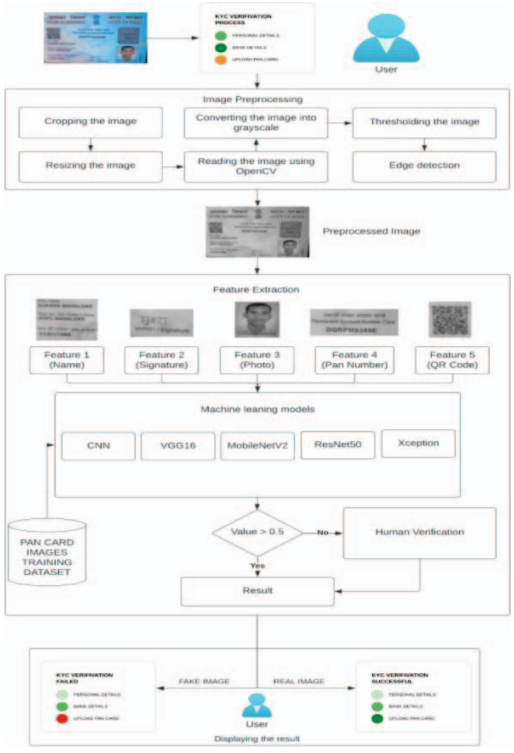


Fig 1. Architecture diagram of the proposed system

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Figure 1 shows the entire architecture of the proposed system which has been implemented. In the initial stage of the process, image preprocessing happens and then followed by feature extraction. The result is then passed to various models like CNN, VGG16, Xception, MobileNetV2 and ResNet50 and each model gives its own result by classifying the image as Real or Fake. The next section talks about the proposed system and the methodology to implement the system.

III. PROPOSED WORK

The PAN card verification process is a comprehensive system designed to ensure the authenticity and integrity of uploaded PAN card images for KYC validation. A web application will be developed and provided to user agencies. These agencies will then authenticate the genuineness of the submitted PAN cards by verifying them through the application. Below is a summary outlining the key steps of the procedure:

1) Collecting diverse PAN card images, annotating for tampering types: A diverse dataset of PAN card images and Aadhar card images are acquired to ensure a representative sample which include variations in lighting conditions, backgrounds, and quality to mimic real-world scenarios. It contains 40 Real PAN card images which were generated using the latest PAN card template available in government websites and 40 fake PAN card images which were generated by tweaking the features of the same template using Canva online image editor tool. In addition, 20 real Aadhar card images and 9 fake Aadhar card images which were generated through the same process were used to analyze the efficiency of the model upon running on different datasets. The current work involves using 40 real cards and an equal number of fake ones. Enhancing the dataset's size and introducing diverse discrePANcies can increase reliability. Also, the web application does not store user data. Therefore, the need for security measures is not considered for this application. However, securing the PAN card generation process is the scope of this future work.

2) Preprocessing images, extracting key features for tampering detection: The collected PAN card images and Aadhar card images undergo thorough preprocessing to standardize their format, resolution, and orientation. CNN preprocessing involves steps like resizing images to a uniform size, normalizing pixel values, and augmenting data through techniques like flipping or rotation. The key steps in feature extraction are image preprocessing, resizing, grayscale conversion and noise reduction. Text recognition extracts data like cardholder's name and PAN number, while numerical data extraction focusses on birth dates. Region analysis spots tampering signs, edge detection highlights card edges and image fusion combines modalities for better detection. These features form a feature vector to distinguish genuine from tampered cards efficiently. Essential features are extracted from these preprocessed images to create a rich dataset that encapsulates the distinctive characteristics associated with tampering. Techniques such as image normalization, cropping, thresholding and edge detection are applied to ensure the model's effectiveness in identifying subtle tampering cues.

3) Developing optimized model for PAN card tampering detection: A tailored CNN model is developed, leveraging the extracted features from the preprocessed images. The model is rigorously fed the dataset and trained meticulously and is equipped to handle real-life scenarios. It is constantly optimized and fine-tuned, which is yet another critical aspect of the optimization process. This entails modifying the pre-trained model's parameters to more closely match the unique properties of PAN card photos and the types of tampering it may undergo. This fine-tuning enhances the model's sensitivity to tampering cues while maintaining a balance to avoid false positives. Overall, the development and optimization of the CNN model involves a strategic combination of advanced neural network architecture, transfer learning, fine-tuning, and careful consideration of sensitivity and specificity.

4) Developing a system for KYC verification model integration with clear documentation: An intricately designed application is created to facilitate the seamless integration of the developed tampering detection model into existing KYC verification processes. This application serves as a bridge between the tampering detection model and various platforms or applications where KYC verifications take place. The documentation generated for the system is a key component of this process. It provides comprehensive guidance for developers and stakeholders, offering a detailed understanding of the system's capabilities, input requirements, and expected output. The documentation not only serves as a technical reference but also includes practical examples and use cases to assist developers in effectively implementing the tampering detection functionality within their KYC workflows.

5) Evaluating model metrics and deploying as a website for integration: The created model is put through a rigorous evaluation process where its performance is measured against a validation dataset using measures including precision, recall, AUC-ROC, specificity, and F1 score. Upon achieving satisfactory results, the model is deployed as a website, making it accessible for integration into diverse platforms and applications. Continuous monitoring is established to ensure optimal functionality and responsiveness.

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6) Iterating model based on feedback and monitoring for emerging techniques: The deployed model is subject to an iterative process based on user feedback and evolving requirements. Regular updates and improvements are made to address any identified shortcomings or adapt to emerging tampering techniques. Continuous monitoring for emerging trends and advancements in tampering methods ensures the model remains resilient and adaptive over time. This iterative approach ensures the sustained effectiveness and relevance of the tampering detection system in a dynamically changing landscape. Hyperparameters settings that control how a model learns, including factors like the learning rate, neural network layers, or decision tree depth have to be fine-tuned to get optimum performance. The next section talks about the algorithms used in the system and their descriptions.

IV. Algorithm Used

CNN, a deep learning algorithm specifically designed for image processing and recognition tasks, will be the one employed for this system. Convolutional, pooling, and fully connected layers are some of the layers that make it up. The image size is (150, 150, 3). There are seven layers which consist of 2 convolution layers, 2 max pooling layers, 1 flattening layer and 2 dense layers. There are a total of 32 filters with kernel size (3, 3) using ReLU as the activation function. The max pooling layer has a size of (2, 2). The first dense layer has 128 neurons and the second layer has 30 neurons.

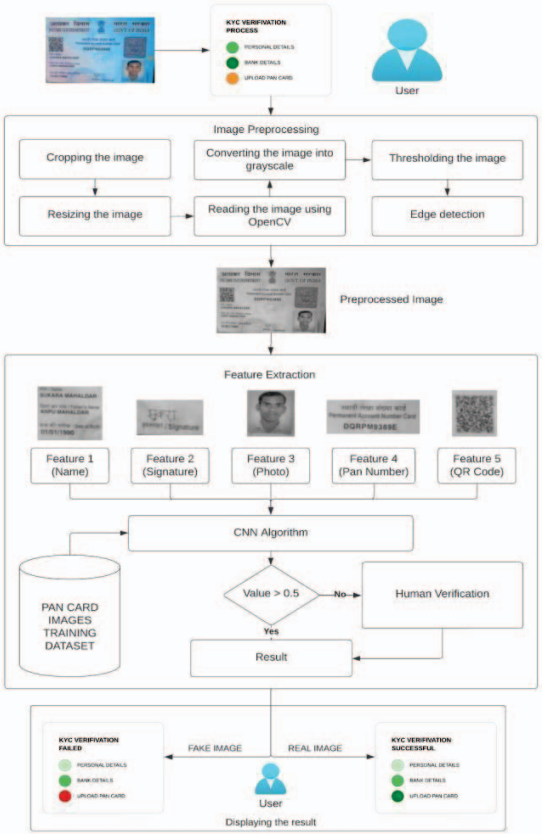


Fig 2 Architecture diagram of the proposed system for CNN

Figure 2 shows the architecture of the CNN model which has been implemented. Preprocessing the images happens in the initial stage of the process where the uploaded image is being cropped, resized and modified into a grayscale image. Then the preprocessed image is passed for feature extraction where various features like Name, Signature, Photo, PAN number and QR code are extracted. These features are passed to the CNN model for classification and the model gives the result as a Real or Fake image based on the classification done by the model.

The PAN card dataset was tested on four other ML models in addition to CNN to check the efficiency of the dataset.

1) VGG16: This is the CNN model that uses 13 convolution and 3 fully connected layers. It is known for its simplicity and effectiveness. The image size is (150, 150, 3). This model consists of 4 layers. The first layer is the base model which is the VGG16 model with 14,714,688 parameters. The second layer is the flattening layer which converts the 4x4x512 feature maps from the base model into a 1D array. The third layer is the first dense layer with 512 neurons that uses the ReLU function as the activation function with 41,94,816 parameters. The fourth layer is the second dense layer with 1 neuron that uses the Sigmoid activation function with 513 parameters.

2) ResNet50: This is a type of CNN model which uses residual blocks in its architecture that introduce skip connections that have been observed to reduce the vanishing gradient problem. Residual blocks consist of multiple convolutional layers, typically with filters of 3x3. The image size is (150, 150, 3). There are a total of 4 layers. The first layer is the base model, the ResNet50 with 23, 587, 712 parameters. The second layer is the flattening layer which converts the 5x5x2048 feature maps from the base model into a 1D array. The third layer is the first dense layer that has 512 neurons with the ReLU activation function and the second fourth layer is the second dense layer with 1 neuron that uses sigmoid function as the activation function.

3) MobileNetV2: It is a CNN model that uses the concepts of residual blocks that helps preserve information flow while using a minimal number of parameters and computations. The image size is (150, 150, 3). There are a total of 4 layers. The first layer is the base model, the MobileNetV2 with 2,257,984 parameters. The second layer is the AveragePooling2D. The third layer is the first dense layer that has 512 neurons with the ReLU activation function and the second fourth layer is the second dense layer with 1 neuron that uses sigmoid function as the activation function.

4) Xception: The spatial and channel-wise convolutions are divided into two distinct processes by Xception using a depth-wise separable convolution algorithm. The image size is (150, 150, 3). There are four layers in this model, the base model layer, the AveragePooling2D layer, the first dense layer which has 512 neurons and uses ReLU as the activation function and the second dense layer which has 1 neuron that uses Sigmoid activation function. The final section talksabout the experimental results and the comparison of

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performance of all models and the performance of the best performing model.

V. Experimental Results

In this study, two different datasets, PAN card and Aadhar card were used to analyze the efficiency of the system. While the model has an accuracy of 100% for PAN card dataset, the CNN model only predicts whether the Aadhar card is authentic or fraudulent with an accuracy of 89% while determining the authenticity of the Aadhar card images. It had a runtime of about 0.24 seconds and with the loss of 0.1405 seconds. This was due to the diversity of the PAN card dataset in comparison with the Aadhar card dataset as the larger volume of diverse images which included changes in name, size of photograph and placement of PAN number yielded in more comprehensive results compared to the Aadhar card dataset which limited the scope of experimentation.

Table 1: Comparing results of different models on an image

MODELS	Accuracy	Loss %	Avg Execution Time (in sec)
CNN	100	0.14	0.24
VGG16	100	0.03	0.4
ResNet50	93.5	0.17	0.41
MobileNetV2	92.5	0.16	0.26
Xception	70.9	0.5	0.31

The performance of the various other models was evaluated on the same PAN card dataset. From the table 1, it is clearly visible that the VGG16 model achieved the highest accuracy of 100% with a runtime of 0.40 seconds. Comparing it to the original tampering detection system of the CNN model, there was only a marginal difference of 0.2%. The ResNet50 model followed with an accuracy of 93.5% and a slightly longer runtime of 0.41 seconds, showing a marginal difference of 0.35% with the base CNN model. The MobileNetV2 model performed with 92.5% accuracy and a faster runtime of 0.26 seconds, with a marginal difference of 0.4% compared to the original CNN model. However, the Xception model lagged with an accuracy of 70.9% and a runtime of 0.31 seconds, showing a marginal difference of 0.25% compared to the original CNN's performance.

Overall, the VGG16 displayed the best accuracy and comparable performance with minimal marginal difference, followed by ResNet50 and MobileNetV2, while Xception showed lower accuracy and slightly longer runtime when compared to the other models. CNN are powerful for preventing image tampering detection because they excel at learning intricate patterns and features within the images, making them highly effective at detecting even subtle alterations or manipulations in the images. Based on the suggested system's performance evaluation, outlines the superiority of performance of CNN compared to other techniques. The proposed CNN parameters are the best option out of all the models that were taken into consideration. As evidenced by its astounding 100% accuracy and 0.14% loss percentage, the CNN model performs remarkably well in correctly identifying data items.

Furthermore, its 0.24 second average execution time demonstrates effective processing power, which is essential for real-time applications where speed is of the essence. CNN outperforms other models in terms of accuracy and execution time, even though other models like VGG16 and ResNet50 also show good accuracy rates. CNN has been shown to be the most consistent and reliable model among those studied, making it the model of choice for jobs requiring accuracy and quick inference.

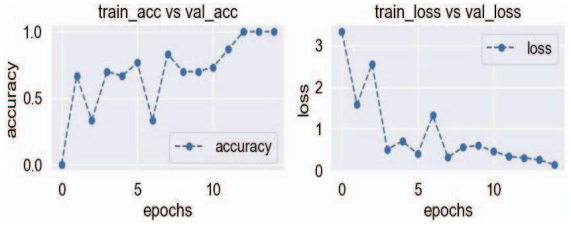


Fig 3. Plot accuracy loss of CNN

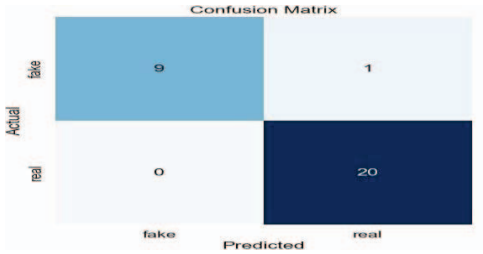


Fig 4. Confusion matrix of CNN

Figure 3 depicts the accuracy of the CNN model and the error loss happened in each iteration of the epochs. This depicts that the accuracy went up to 100% and the error loss came down to 0.14% upon completion of the 15 iterations. Figure 4 depicts the classes which the CNN model has classified for the testing data set. The CNN model has classified 20 images as True Positive, 9 images as True Negative and 1 image as False Positive and 0 images as False Negative.

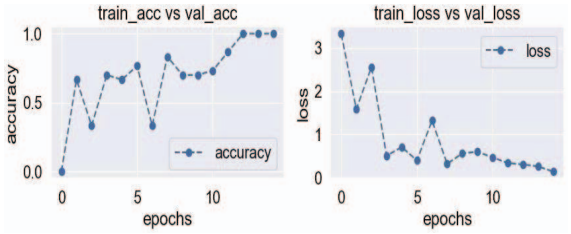


Fig 5 Plot accuracy loss of Xception

Figure 5 depicts the accuracy of the Xception model and the error loss happened in each iteration of the epochs. This depicts that the accuracy went up to 70.9% and the error loss came down to 0.5% upon completion of the 15 iterations. Figure 6 depicts the classes which the Xception model has classified for the testing data set. The Xception model has

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classified 0 images as True Positive, 11 images as True Negative and 0 images as False Positive and 20 images as False Negative.

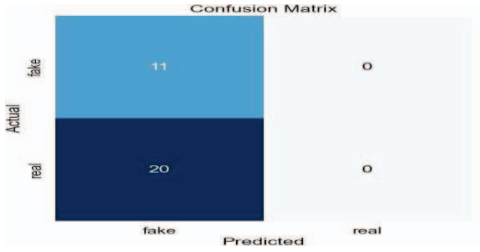


Fig 6. Confusion matrix of Xception

Figure 7 depicts the accuracy of the VGG16 model, and the error loss happened in each iteration of the epochs. This depicts that the accuracy went up to 100% and the error loss came down to 0.03% upon completion of the 15 iterations. Figure 8 depicts the classes which the VGG16 model has classified for the testing data set. The VGG16 model has classified 0 images as True Positive, 10 images as True Negative and 0 images as False Positive and 19 images as False Negative.

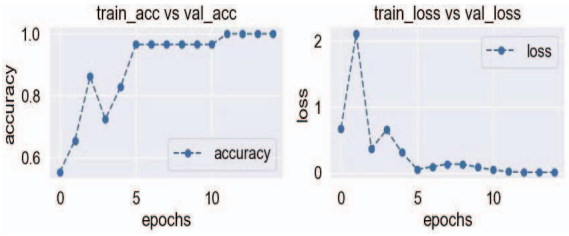


Fig 7. Plot accuracy loss of VGG16

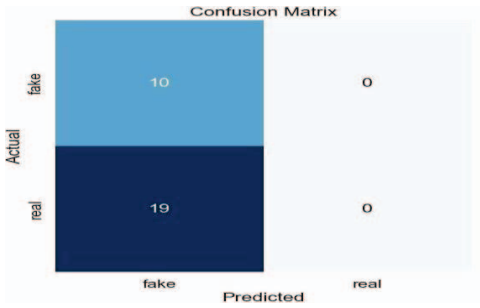


Fig 8. Confusion matrix of VGG16

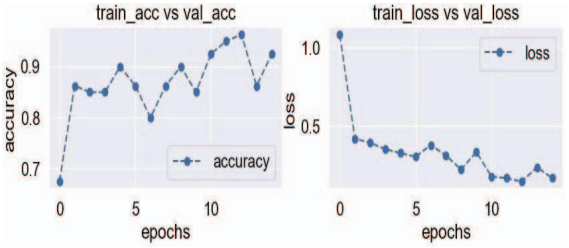


Fig 9. Plot accuracy loss of MobileNetV2

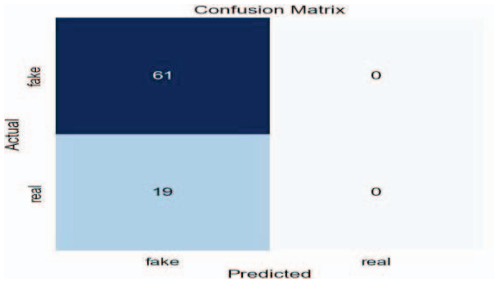


Fig 10. Confusion matrix of MobileNetV2

Figure 9 depicts the accuracy of the MobileNetV2 model and the error loss happened in each iteration of the epochs. This depicts that the accuracy went up to 92.5% and the error loss came down to 0.16% upon completion of the 15 iterations. Figure 10 depicts the classes which the MobileNetV2 model has classified for the testing data set. The MobileNetV2 model has classified 0 images as True Positive, 61 images as True Negative and 0 images as False Positive and 19 images as False Negative.

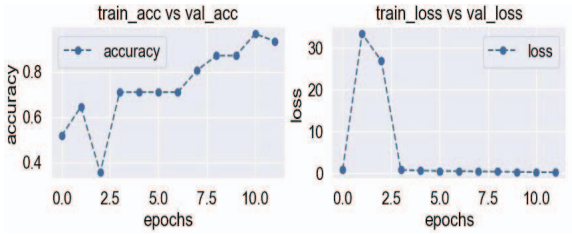


Fig 11. Plot accuracy loss of ResNet50

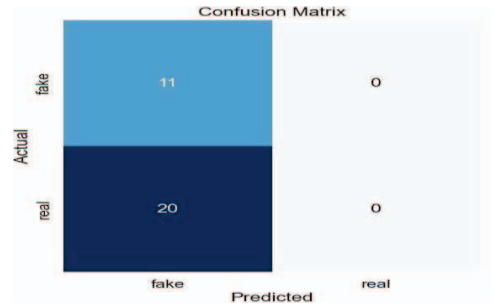


Fig 12. Confusion matrix of ResNet50

Figure 11 depicts the accuracy of the Xception model, and the error loss happened in each iteration of the epochs. This depicts that the accuracy went up to 93.5% and the error loss came down to 0.17% upon completion of the 15 iterations. Figure 12 depicts the classes which the Xception model has classified for the testing data set. The Xception model has classified 0 images as True Positive, 11 images as True Negative and 0 images as False Positive and 20 images as False Negative.

VI. Conclusion

In conclusion, the Fake Card Detection system with the integration of CNN provides a proactive approach to fight against identity theft and financial fraud. The model,

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empowered by the training and testing using both PAN card and Aadhar card datasets underwent rigorous training, testing and analysis using the models VGG16, ResNet50, MobileNetV2 and Xception and proved its robustness and accuracy with which it determines if the given image is fake or not along with a very precise determination of the tampering percentage. Through the training and the classification phases, the model can identify the slight changes made in the cards and find the differences between the genuine and fake cards, thereby improving its capacity in securing the financial transactions and verification of identities. From the ensemble of models used, CNN has been the best choice and has shown excellent results in terms of accuracy,robustness, runtime and computational efficiency. Besides, aversatile inability to adapt to different sizes and complexities of images makes it suitable for real-time applications, where manipulation techniques in the documents are common and vary with the quality of the document used. In addition, the combination of both the PAN and Aadhar card datasets extends the system's reach by providing cross-validation and additional insights towards individual identity verification. This combination not only increases the system's accuracy intamper detection but also increases its resilience againstsophisticated fraud techniques that may involve manipulation across multiple identification documents. The proposed PAN card tampering detection system helps to identity fraud and financial crimes. The integration of cutting-edge technology along with accuracy and precision increases the trust that the users have in the proposed system. As technology continues to evolve, this system serves as a milestone in proactive measures to ensure individual privacy, security, and financial welfare in a highly digital world.

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